Age Group Estimation

ITCS352 DEEP LEARNING

Group 10

6688040 Sippakorn Kanin
6688052 Pannawit Krutnak
6688063 Chawin Marom

Problem Definition

The Problem:

Age estimation from facial images is challenging due to non-linear aging patterns and high variation between individuals, making it difficult to train a reliable classifier without careful dataset and model selection.

Motivation:

Understanding which transfer learning strategy — head-only vs. two-phase fine-tuning — best adapts pretrained CNNs to a four-class age group classification task (18–24, 25–39, 40–59, 60+) on UTKFace.

Impact:

Two-phase fine-tuning with differential learning rates, class-weighted loss, and cosine annealing consistently outperforms head-only transfer learning, with the best model achieving 68.7% test accuracy deployed as a real-time Streamlit web application.

Challenge:

Class imbalance in UTKFace (25–39 group nearly 3x larger than minority classes), overfitting during full fine-tuning, and selecting appropriate learning rates per layer group to preserve pretrained features while adapting to the task.

Dataset Description

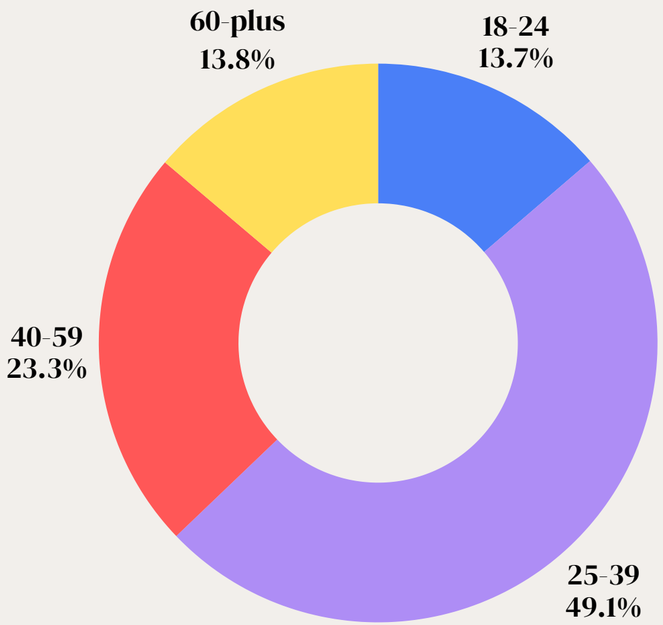
Data source:
Kaggle UTKFace Dataset with 20k+ pictures with the range of 0-116 years old with labels of age, gender, ethicnity.

Size: 200x200 pixels

Issues: Class Imbalance, Small picture resolution.

A total of 19,475 pictures
Class distribution:

- 18-24: 2,670 pictures
- 25-39: 9,571 pictures
- 40-59: 4,544 pictures
- 60-plus: 2,690 pictures



UTKFace dataset

24 years old

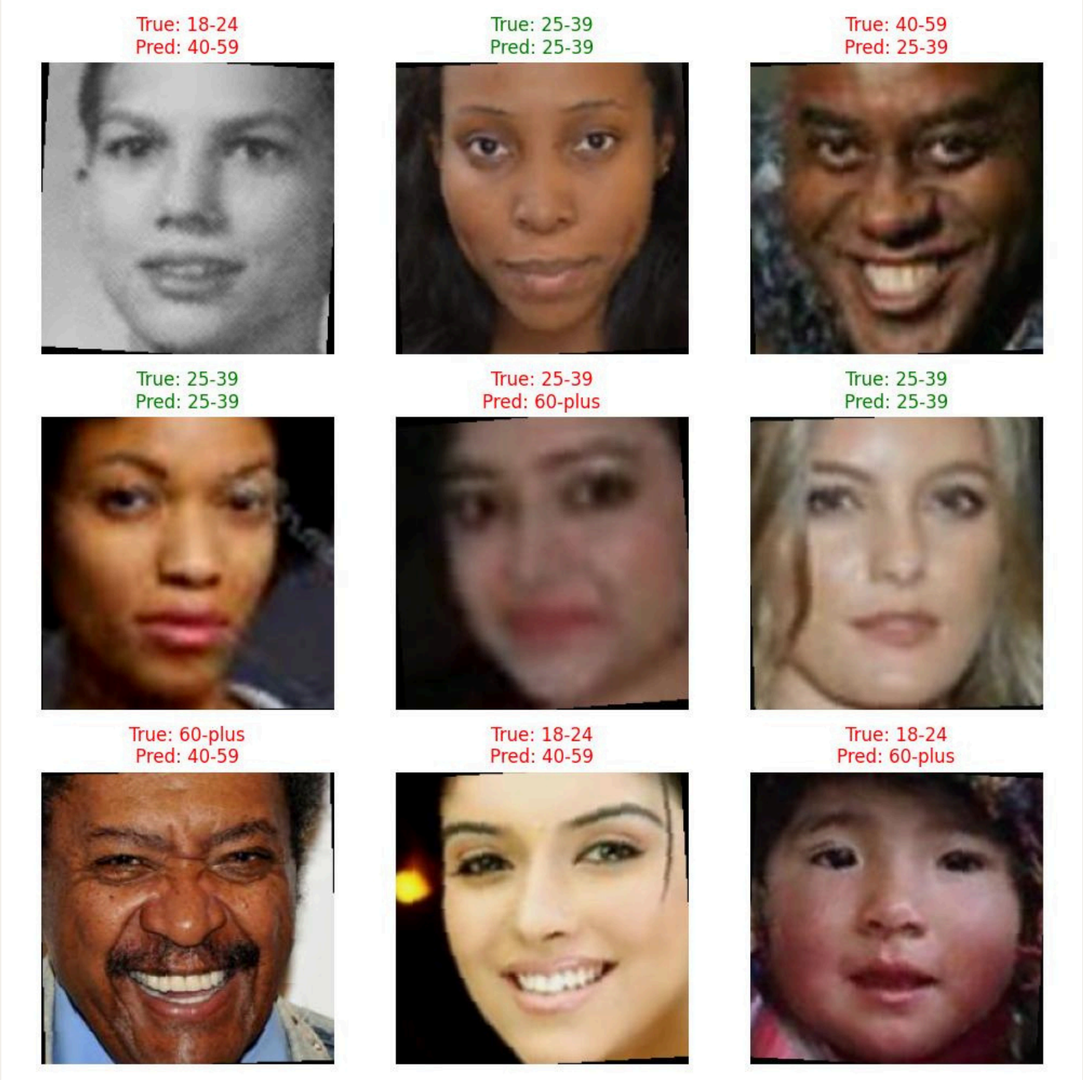
75 years old

Preprocessing

- Resizing to 224x224 pixels
- Normalization using ImageNet mean of ([0.485, 0.456, 0.406])
- Normalization using standard deviation of ([0.229, 0.224, 0.225])

Augmentation

- Random Horizontal flip
- Random Rotation of 5 degrees
- Color jitter (Brightness and Contrast)



Dataset after preprocessing and augmentation

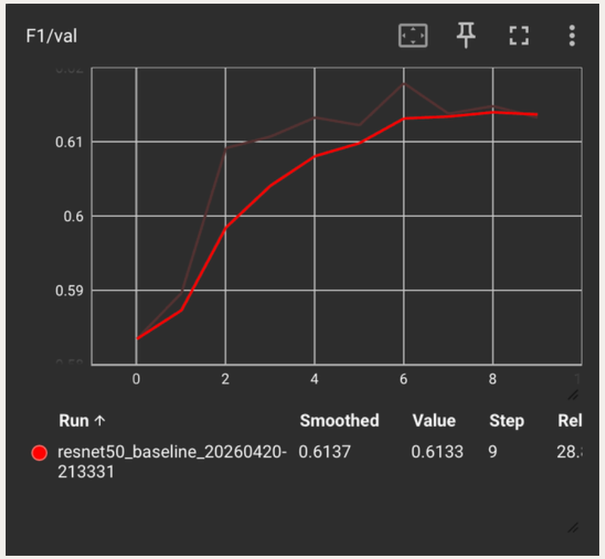
Model development

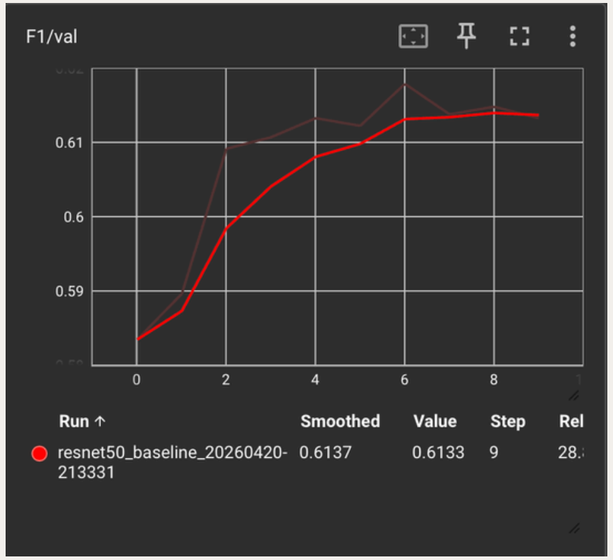
Baseline model

Pretrained ResNet-50 with fully frozen backbone. Only the final linear classification head was trained for 10 epochs using Adam optimizer. Achieved **60.4% test accuracy** and **60.4% macro F1.**

Problems

The frozen backbone cannot adapt its feature representations to the age group task. The 25–39 class dominates predictions due to class imbalance, and the simple linear head lacks regularization, leading to mild overfitting





Model development

Improvement: Finetuned ResNet-50

Train the model for 2 phases using **15** epochs in the first phase, training only the classification head with backbone frozen.

60 epochs with **15** patience early stopping in the second phase training the backbone layer with different learning rate for each layer.

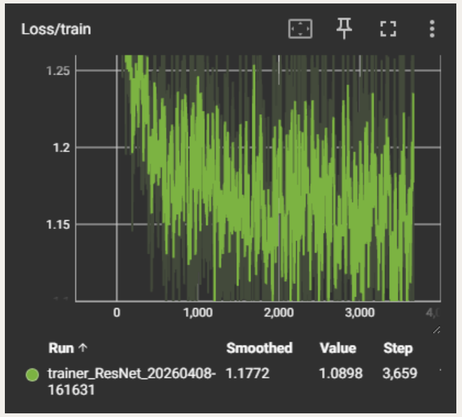
Adding **class weight** to control how much the model learn from each age group and **label smoothing** to prevent the model from overfitting and improving its performance.

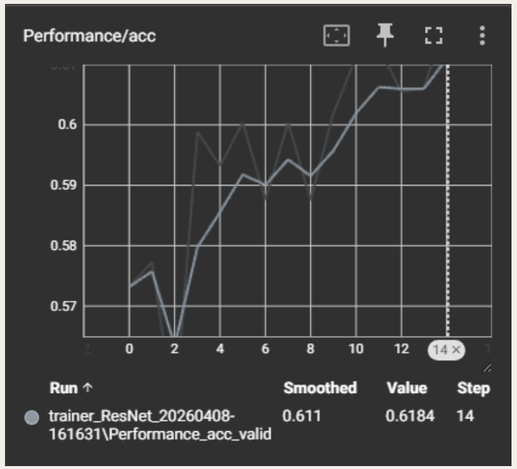
```
class_weights = torch.tensor([1.89, 1.0, 1.45, 1.88]).to(device)
loss_fn = nn.CrossEntropyLoss(weight=class_weights, label_smoothing=0.1)
```

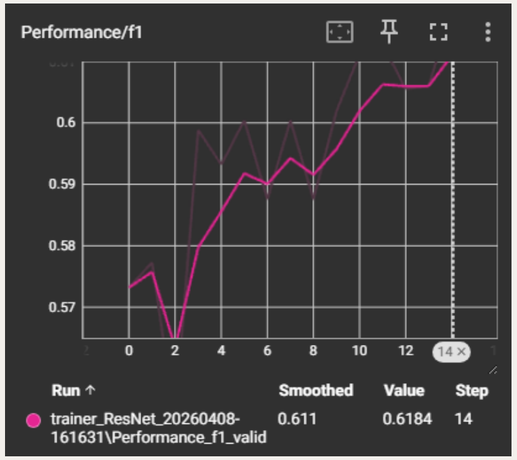
Comparing models

The model got better with the test accuracy and macro F1 of **64.6%**

Comparing to the baseline which achieved **60.4%** test accuracy and macro F1







Model development

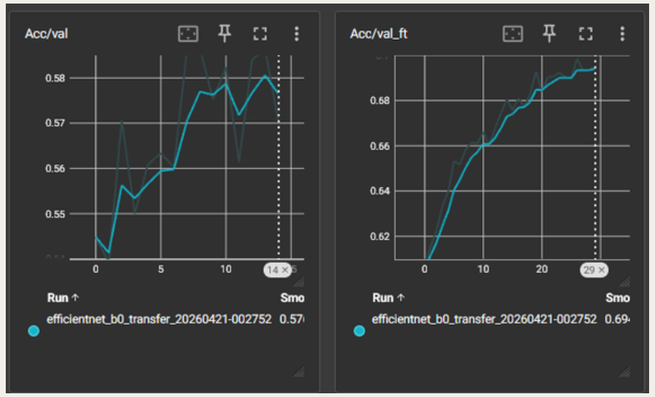
EfficientNet-B0

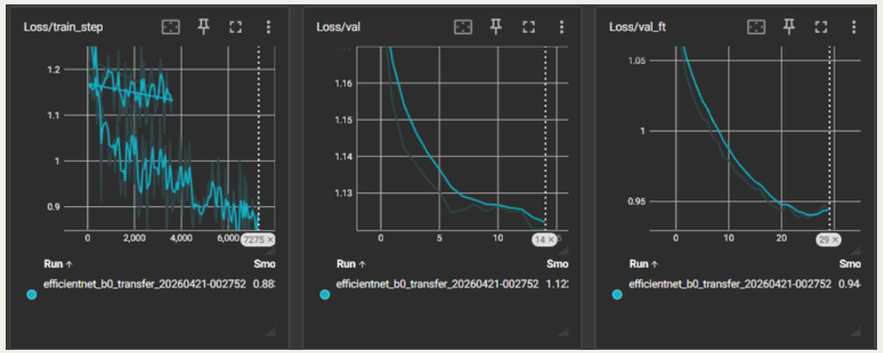
- Lightweight, parameter-efficient backbone
- Highly suitable for real-time inference
- Classification head: Dropout(0.5) + Linear

Achieved 68.7% test accuracy and 68.7% macro F1.

Two-Phase Training

- Phase 1 (15 ep.): frozen backbone, head training
- Phase 2 (30 ep.): full unfreeze
  - Backbone LR: 10−5
  - Head LR: 10−4
  - Weight decay: 10−2





Strategies for Robust Training

Preventing Overfitting

- **Dropout (0.5)** on classification head provides explicit regularization
- **Data augmentation** increases training distribution diversity
- **Early stopping** (patience=15) Stop the training when the validation loss reaches a constant level.

Optimizing Learning Dynamics

- **Different Learning Speeds:** We train the older parts slowly to keep basic knowledge, and the new parts faster to adapt quickly.
- **Smooth Slow-downs & Quick Boosts:** We slow the learning to focus on details, with occasional boosts to avoid getting stuck.
- **Stable and Reliable Results:** This helps the model find a stable answer, making it reliable in real-world use.

Performance Comparison

	Test Accuracy	Test F1-score
ResNet-50 Baseline	60.4%	60.4%
ResNet-50 Finetuned	64.6%	64.6%
EfficientNet-B0	68.7%	68.7%

Trade-off

Baseline model

- Short training time, low accuracy and performance.

Finetuned model

- Longer training time, higher accuracy, less overfitting.

EfficientNet

- Lightweight model, Higher accuracy with less training time.

Live Demo

Docker